

MECHANICAL ENGINEERING MATERIALS*Time Allowed: 2.5 Hours**Full Marks: 60*

Answer to Question No. 1 of Group A is compulsory and to be answered first. This answer is to be made in separate loose script(s) provided for the purpose. Maximum time allowed is 30 minutes, after which the loose answer scripts will be collected and fresh answer scripts for answering the remaining part of the question will be provided. On early submission of answer scripts of Question No. 1, a student will get the remaining script earlier.

Answer any Five (05) Questions from Group B.

Group A

1. Choose the correct answer from the given alternatives (any twenty): (1x20)
 - i) Coordination number FCC structure is - (a) 6 (b) 8 (c) 10 (d) 12.
 - ii) Pure iron at room temperature has the following crystal structure. - (a) BCC (b) FCC (c) HCP (d) SC.
 - iii) Atomic packing factor of HCP structure is - (a) 0.74 (b) 0.68 (c) 0.52 (d) none of these.
 - iv) The property by virtue of it a material can be rolled into thin sheet is called a) Ductility b) elasticity c) malleability d) Fatigue.
 - v) The continuous deformation under steady load is called - (a) creep (b) fatigue (c) corrosion (d) none of these.
 - vi) Hardest micro structure of steel is - (a) Ferrite (b) Martensite (c) Bainite (d) Pearlite.
 - vii) The steel having more than 0.8% carbon are called - (a) eutectoid steels (b) hypo eutectoid steel (c) hyper eutectoid steel (d) none of these.
 - viii) The property of material by virtue of which it undergoes permanent deformation with the application of a load, is called - (a) hardness b) plasticity c) Fatigue d) elasticity.
 - ix) In Annealing heat treatment cooling is done in (a) air (b) Furnace (c) water (d) None of these
 - x) Product of blast furnace is - (a) Wrought Iron (b) Cast Iron (c) Pig Iron (d) Steel
 - xi) In low carbon steel C % varies from - (a) 0.008-0.3 (b) 0.3-0.6 (c) 0.6-2.0 (d) 2.0-4.3
 - xii) Cast iron is a - (a) ductile material b) malleable material c) brittle material d) tough material.
 - xiii) Which of the following iron exist at room temperature? - (a) alpha Iron (b) Beta Iron (c) gamma Iron (d) delta Iron
 - xiv) The hardness of steel increases if % of carbon (a) decreases (b) Increases (c) all of the above (d) None of the above
 - xv) Tempering heat treatment is done after - (a) Hardening, (b) normalising, (c) Annealing, (d) None of the above.
 - xvi) Which is surface hardening process? a) Annealing b) Normalising c) Nitriding d) hardening
 - xvii) The lower critical point for all steel is- (a) 600°C (b) 723°C (c) 700°C (d) 913°C.
 - xviii) Brass is an alloy of - (a) copper and zinc (b) copper and tin (c) copper, Aluminium (d) none of these.

- xix) Duralumin is an alloy of – a) Aluminium b) iron c) Nickel d) None of the above
- xx) Babbit metal is used in – (a) Bearing (b) Gear (c) Shaft (d) None of these
- xxi) Bottles from thermoplastic materials are made by – (a) Compression moulding b) Extrusion moulding c) Injection moulding d) blow moulding.
- xxii) Long plastic pipes are produced by – (a) Compression moulding b) extrusion moulding c) Injection moulding d) blow moulding.
- xxiii) Examples of Thermosetting plastics are – (a) Electric switch (b) Pressure cooker handle c) all of the above d) None of the above
- xxiv) Example of non destructive test is- (a) ultrasonic test (b) radiographic test (c) magnetic particle test (d) all of these.
- xxv) Non- destructive testing is used to determine a) Location of defects, b) Chemical composition, c) Corrosion of metal, d) All of these.

Group B

2. a) Write down the names of different types of crystal structures in metals. (2)
b) Write down the number atoms per unit cell, co ordination number, atomic packing factors and relation between sides of unit cell and atomic radius of simple cubic and BCC structure. (6)
3. a) Define hardness, Brittleness, ductility and Elasticity. (4)
b) Draw schematic diagram showing flow path for making steel and cast iron, wrought iron in different furnaces from iron ore. (4)
4. a) Draw the cooling curve for pure iron showing salient points. (2)
b) Draw Iron carbon equilibrium diagram up to 2 % carbon showing micro constituent present in (4)
c) Classify steel according to % of carbon. <https://www.wbscteonline.com> (2)
5. a) Explain the various purpose of heat treatment. (3)
b) Describe hardening heat treatment process and also write down its application. (5)
6. a) Write short note on any one i) Nitriding ii) Flame hardening iii) Austempering (4)
b) Draw the TTT diagram for eutectoid steel showing salient points. (4)
7. a) Write down the names of two copper based alloys. (2)
b) What are the properties of aluminium alloys? (2)
c) Describe briefly the properties and application of bearing metals. (4)
8. a) Compare Thermoplastics and Thermosetting plastic. (3)
b) Describe Compression moulding or Injection moulding process with the help of simple sketch. (5)
9. Write short notes on any two i) Fibre reinforced Composite Material ii) Ceramics iii) Bakelite
iv) refractories v) polyester (4+4)
10. a) Write down the names of different Non destructive testing methods. (2)
b) What are the differences between destructive and non destructive testing methods? (2)
c) Describe any one non destructive testing method with suitable sketch. (4)